

Exam 1 Cheat Sheet

Data Types: Quantitative = numeric (Discrete = countable; Continuous = measured). Qualitative = categories (Nominal = no order; Ordinal = ordered). **Parameter** = describes whole population. **Statistic** = describes a sample. **Sample vs Population:** Use a *sample* when population is too large to measure (e.g. avg weight of all salmon). Use *population* when data is already small/complete (e.g. avg grade of one class). **Key vocab:** Individual = one member. Variable = what is measured. Population = whole group. Sample = subset. Parameter = number describing population. Statistic = number describing sample. **Measurement Levels:** Nominal → Ordinal → Interval (no true zero, e.g. temp °F) → Ratio (true zero, e.g. weight).

Experiment = researcher *assigns* treatment. **Observational** = just watches (Cross-sectional = one time; Case-control = compare groups backward; Cohort = follow forward). **Randomized** = researcher randomly assigns treatment. **Matched pairs** = same variable on same/matched objects. **Blind** = subject doesn't know treatment. **Double-blind** = neither knows. **Neither** = both know. **Sampling bias** = unequal selection chances. **Stratified** = sample proportionally from subgroups. **With replacement** = item returned to pool.

Probability

- **Outcome** = one possible result. **Event** = collection of outcomes. **Impossible** → $P=0$. **Certain** → $P=1$.
- **Joint probability** = probability at intersection of a row and column in a contingency table.
- **Complement:** $P(\text{not } A) = 1 - P(A)$ ← *complement event* = when A does NOT occur.
- **Addition Rule:** $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$
- **Mult. Rule (independent):** $P(A \text{ and } B) = P(A) \cdot P(B)$
- **Conditional:** $P(A|B) = P(A \text{ and } B) \div P(B)$ ← restrict to "given" group
- **Unfair/biased coin** = heads and tails are NOT equally likely.
- **Counting:** Sample space size = multiply outcomes of each step (e.g. $6^5 \times 2^3$ for 5 dice + 3 coins).

Random Variables & Expected Value

- **Expected Value (μ):** $\mu = \sum [x \cdot P(x)]$ — long-run average of a random variable.
- **Binomial RV** = counts successes in n independent Bernoulli trials; $P(S)=p$, $P(F)=1-p$.
- **$P(X=k) = C(n,k) \cdot p^k \cdot (1-p)^{n-k}$** , $C(n,k) = n!/[k!(n-k)!]$
- **$P(a < X < b)$** = sum probs strictly between a and b ; **$P(X > a)$** = $1 - P(X \leq a)$.

Poisson Distribution (counts of rare events over time/space)

$P(X=k) = (e^{-\lambda} \cdot \lambda^k) / k!$, where λ = mean rate, $e \approx 2.71828$

- $P(X < n) = \text{sum } P(X=0) + P(X=1) + \dots + P(X=n-1)$
- $P(X > n) = 1 - P(X \leq n)$

Exponential Distribution (time *between* Poisson events)

$X \sim \text{Exp}(\lambda)$: $\mu = 1/\lambda$ (mean time between events) $P(X \leq x) = 1 - e^{(-\lambda x)}$ $P(a < X < b) = P(X \leq b) - P(X \leq a) = e^{(-\lambda a)} - e^{(-\lambda b)}$

Graphs & Distribution Shapes

Data	Graph
1 categorical var	Bar chart / Pie chart
1 quantitative var	Histogram / Stem-leaf / Dot plot
2 quantitative vars	Scatterplot
2 categorical vars	Contingency table

Distribution shapes: Symmetric/Bell = balanced. Skewed right = tail pulls right (few high values). Skewed left = tail pulls left (few low values). Uniform = all bars roughly equal height.

- **Relative Frequency** = frequency $\div$ total. **Class width** = upper limit - lower limit.
- **Stem-and-leaf:** stem = all but last digit; leaf = last digit, sorted, comma-separated; no data $\rightarrow$ "DNE".
- **Contingency table:** joint prob = cell $\div$ grand total; conditional prob = cell $\div$ row or column total.

Quick Decision Trees

Study type: Researcher assigns treatment? Yes $\rightarrow$ Experiment / No $\rightarrow$ Observational **Blinding:**

Neither knows $\rightarrow$ Double-blind; Only subject doesn't know $\rightarrow$ Blind; Both know $\rightarrow$ Neither

Observational timing: One point $\rightarrow$ Cross-sectional; Compare groups backward $\rightarrow$ Case-control;

Follow forward $\rightarrow$ Cohort **Sample vs Population:** Can you realistically measure everyone?

No $\rightarrow$ Sample / Yes $\rightarrow$ Population